

Murat Khidoyatov

U.S. Permanent Resident

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Education

Yale University

Graduation Date: May 2027

B.S. Mechanical Engineering (ABET)

Relevant Coursework: Statics & Dynamics, Mechanical Design, Material Science, Thermodynamics, Fluid Mechanics, Programming in C++, Electrical Engineering

Awards: Awarded a competitive Yale personal research grant (8% acceptance rate).

Technical Skills

Design & Modeling: SolidWorks, OnShape, GD&T (ASME Y14.5), DFM, DFA

Simulation & Analysis: ABAQUS, COMSOL, FEA, CFD, MATLAB, V&V

Fabrication: Machine Shop Tools, 3D Printing (SLA, FDM), Sheet Metal Fabrication

Programming: Python, C/C++, JAX, PyTorch

Work Experience

Oak Ridge National Laboratory

Oak Ridge, TN · Summer 2025

Mechanical Engineer R&D Intern

- Designed and fabricated 3D-printed tooling for nuclear material handling using DFM principles, reducing process cycle time by 66% (12+ hr to 4 hr).
- Performed hardness and conductivity characterization on 100+ aluminum alloy samples to develop processing maps, gaining hands-on experience with structural material testing methods.
- Developed ML-based pipeline to predict alloy material properties within 4% accuracy for EV manufacturing applications.

Technical Engineering Experience

Yale Undergraduate Robotics

2024 – Present

President & Rover Sub-team Lead

- Led mechanism design and structural integration of a deployable Mars rover life-detection system, managing a 10-member team with full BOM and part drawing documentation.
- Validated a 50N linear actuator deployment mechanism to a 2.0x safety factor using FEA (COMSOL) and physical load testing.
- Designed and validated a 4-DoF robotic arm by developing a custom forward/inverse kinematic simulator for path planning and joint interference analysis.
- Optimized actuator structural rigidity and optical alignment using SolidWorks and COMSOL FEA; fabricated custom sheet metal extrusions and integrated embedded sensor systems.

Yale – AID

2023 – 2025

Glove Project Team Lead & Vice-President

- Led a 7-member multidisciplinary team to deliver a custom passive prosthetic limb to a pediatric patient, securing \$5,000+ in project funding.
- Performed root cause analysis on pneumatic actuator failures and iterated on 3D-printed silicone mold designs to improve mechanism reliability.
- Optimized prosthetic stiffness and flexibility via SolidWorks FEA; applied DFA principles for ease of assembly and maintenance.

Yale Engineering Humphrey Lab

New Haven, CT · 2023 – Present

Undergraduate Researcher

- Developed a Physics-Informed Neural Network (PINN) simulating elastic deformation of hyperelastic materials within 7% accuracy vs. COMSOL/ABAQUS benchmarks, supporting materials selection for flexible aerospace structures.
- Architected a deep learning model in JAX/PyTorch to automate constitutive equation discovery for elastomers.